RISK 201: Rates of Return

Himchan Jeong, FSA

Department of Statistics & Actuarial Science
Simon Fraser University

7.1 Net present value

Notation:

For $t = 0, 1, \ldots, n$,

- A_t: Amount received at time t from an investment
- B_t: Amount paid into an investment at time t
- $C_t = A_t - B_t$: net amount received at time t from an investment.

Usually, we have $C_0 < 0$ and $C_n > 0$.

$ paid in $ received

Net present value (NPV)

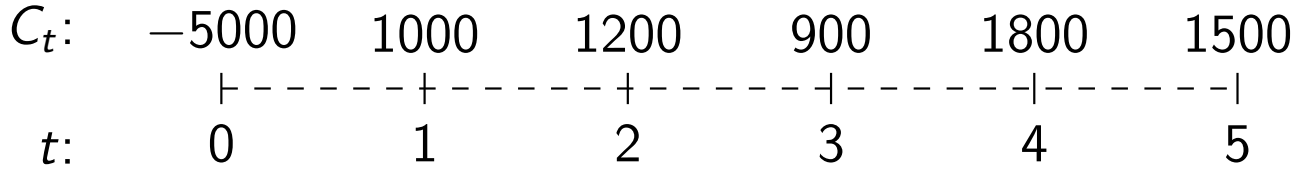
Let $P(i)$ denote the net present value of an investment, valued at rate i:

$$P(i) = \sum_{t=0}^{n} C_t \, v^t$$

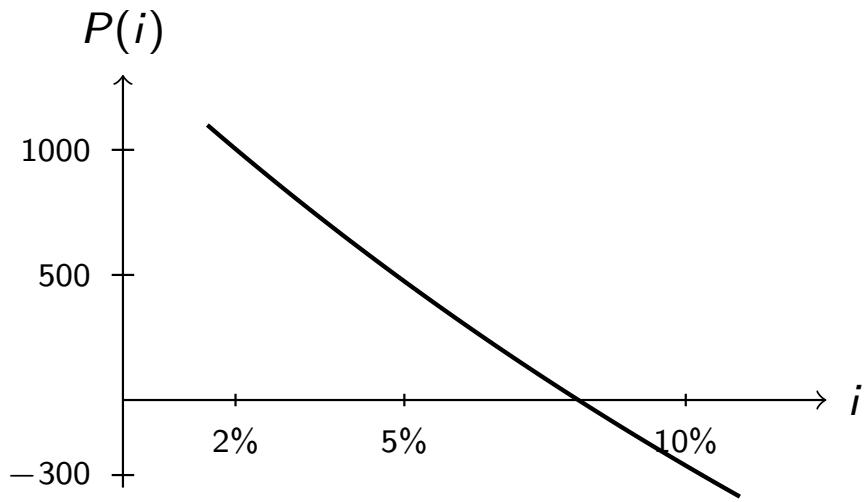
The NPV is generally a decreasing function of i.

Example 7.1

Consider the following series of cash flows:



i	$P(i)$
2%	1003.40
5%	474.42
10%	−262.18



Discounted cash flow analysis

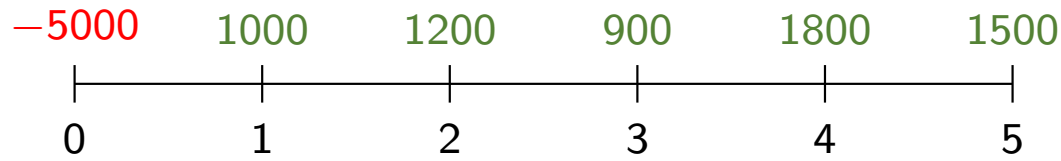
- Every investor is assumed to have an investment preference rate (i_p) which they would like to earn on their investments.
- An investment is acceptable if its NPV is positive when valued at the investor's preference rate:

$$P(i_p) = \sum_{t=0}^{n} C_t \frac{1}{(1+i_p)^t} > 0$$

- Normally, a higher i_p makes an investment less desirable.

Example 7.1, continued

The cash flows are:

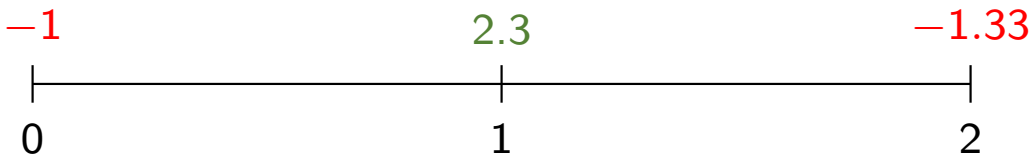


If $i_p = 5\%$, $P(i_p) =$

If $i_p = 10\%$, $P(i_p) =$

Example 7.2

Consider the following stream of cash flows.



If $i_p = 5\%$, $P(i_p) =$

If $i_p = 15\%$, $P(i_p) =$

7.2 Internal rate of return

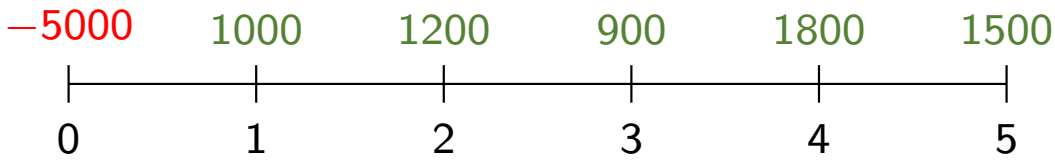
The **internal rate of return (IRR)** is the interest rate i which satisfies

$$P(i) = \sum_{t=0}^{n} C_t \frac{1}{(1+i)^t} = 0.$$

An investor prefers a higher IRR; a borrower prefers a lower IRR.

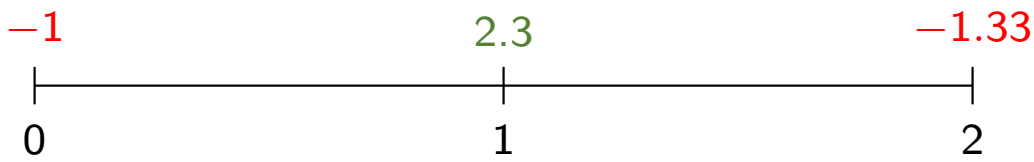
$\text{IRR} > 0$	the investment grows in value
$-1 < \text{IRR} < 0$	some of the original investment is lost
$\text{IRR} < -1$	more than original investment is lost

Example 7.1, continued



Confirm that the IRR is approximately 8.1%.

Example 7.2, continued



Uniqueness of IRR

- The IRR need not be positive, exist or be unique.
- Descartes' rule of signs:
  - The **maximum number of positive roots** of an nth degree polynomial is equal to the **number of sign changes in the coefficients**.
  - If all the cash flows in one direction (i.e., $+$ or $-$) are made before the cash flows in the other direction, then the IRR is unique (because in this case, the number of sign changes $= 1$).

IRR method of capital budgeting

1. Calculate the IRR of the investment
2. Compare this IRR to the interest preference rate (i_p) of the investor:
   - if IRR $>$ i_p, consider investment
   - if IRR $<$ i_p, reject investment

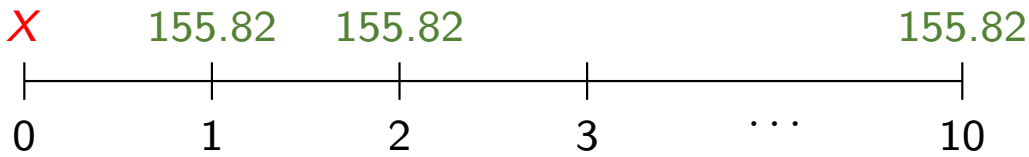
IRR method of pricing

1. Consider the investor's preference rate i_p.
2. Set the price of the investment so that the IRR is equal to i_p. This is equivalent to setting the NPV valued at i_p to zero.

$$P(i_p) = \sum_{t=0}^{n} C_t \frac{1}{(1+i_p)^t} = 0$$

Example 7.3

Consider an annuity that pays 155.82 at the end of each year for 10 years. If the investor's preference rate is 3% per year, find the maximum price she is willing to pay for this annuity.



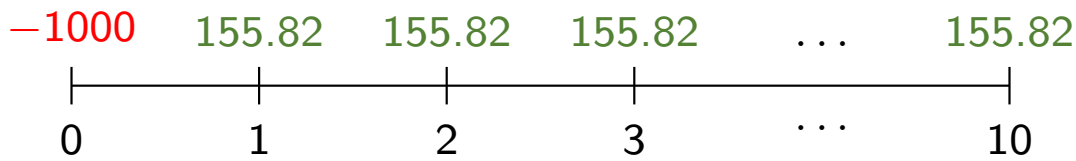
Borrowing to invest

Suppose you are considering an investment opportunity but you don't actually have funds to invest $\rightarrow$ you can borrow money.

- The loan you receive should have the same term (length) as the investment.
- You would only consider an investment if you could make a profit after taking into account the cost of the loan (i.e., interest on the amount you borrowed).

Example 7.4

Consider the following investment:



Suppose you have to borrow the initial \$1000 needed to invest. If you can borrow at $i = 10\%$, should you go ahead?

Q: Would the cash inflows from this investment cover the cost of borrowing at $i = 10\%$?

Example 7.4, continued

Period	Payments	Interest	Principal	Loan balance
0				1,000.00
1	155.82	100.00	55.82	944.18
2	155.82	94.42	61.40	882.78
3	155.82	88.28	67.54	815.24
...	...	...	...	...
10	155.82	24.20	131.62	110.37

Example 7.4, continued

If the cost of borrowing is 10%, borrowing to invest is not profitable: cash inflows from investment are not enough to pay off the loan you took out to invest!

In present value terms, at a rate of $i = 10\%$:

$$\sum_{t=0}^{n} A_t \frac{1}{(1+i)^t} < \sum_{t=0}^{n} B_t \frac{1}{(1+i)^t}$$

$$P(0.1) = \sum_{t=0}^{n} A_t \frac{1}{(1.1)^t} - \sum_{t=0}^{n} B_t \frac{1}{(1.1)^t} = -42.55$$

Example 7.4, continued

Would the cash inflows from this investment cover the cost of borrowing if you could borrow at 5%?

Period	Payments	Interest	Principal	Loan balance
0				1,000.00
1	155.82	50.00	105.82	894.18
2	155.82	44.71	111.11	783.07
3	155.82	39.15	116.67	666.40
...	...	...	...	...
10	155.82	−8.43	164.16	−330.99

Example 7.4, continued

If the cost of borrowing is 5%, borrowing to invest is profitable: cash inflows from investment are more than enough to pay off the loan you took out!

In present value terms, at a rate of $i = 5\%$:

$$\sum_{t=0}^{n} A_t \frac{1}{(1+i)^t} > \sum_{t=0}^{n} B_t \frac{1}{(1+i)^t}$$

$$P(0.05) = \sum_{t=0}^{n} A_t \frac{1}{(1.05)^t} - \sum_{t=0}^{n} B_t \frac{1}{(1.05)^t} = 203.20$$

Example 7.4, continued

What is the maximum loan rate that would still allow you to invest without losing money?

$\rightarrow$ Find i such that

$$P(i) = \sum_{t=0}^{n} A_t \frac{1}{(1+i)^t} - \sum_{t=0}^{n} B_t \frac{1}{(1+i)^t} = 0$$

IRR, NPV, and the borrowing criterion

By definition, the PV of cash inflows is exactly equal to the PV of cash outflows when valued at the IRR:

$$\sum_{t=0}^{n} A_t \frac{1}{(1+IRR)^t} = \sum_{t=0}^{n} B_t \frac{1}{(1+IRR)^t}$$

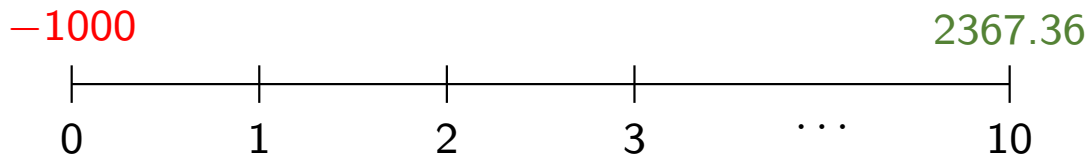
The IRR is the loan rate that would allow you to break even if you had to borrow the funds to invest.

The net present value $P(i)$ is the PV of the profit you would make on your investment, after taking into account the cost of borrowing at rate i.

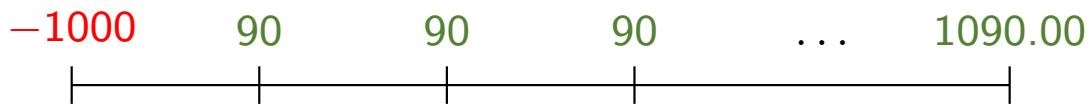
Example 7.5

Find the IRR of the following investments:

(a)

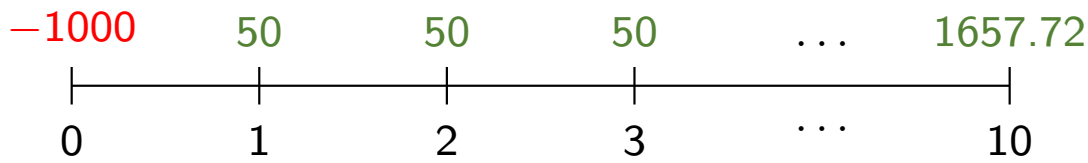


(b)

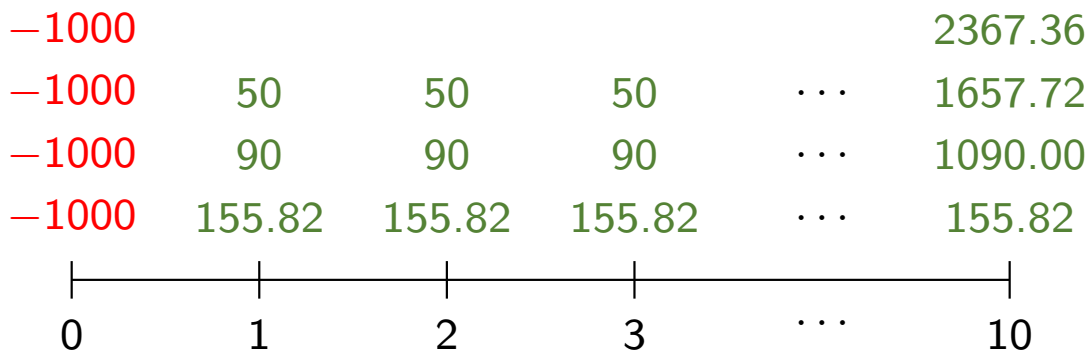


Example 7.5, continued

(c)



Example 7.5, continued



Taking into account reinvestment

A reinvestment rate is the rate at which a payment received from an original investment can be reinvested.

When the proceeds of an investment are reinvested, the appropriate focal date for the valuation of cash inflows and outflows is at the **end** of the investment horizon.

$$\Rightarrow \text{compare AVs instead of PVs}$$

Yield rate

A measure of the rate of return on an investment that takes reinvestment into account is the yield rate.

The yield rate i satisfies

$$\sum_{t=0}^{n} A_t (1+j)^{n-t} = \sum_{t=0}^{n} B_t (1+i)^{n-t}$$

where j is the reinvestment rate.

Yield rate and the borrowing criterion

You can think of the yield rate as the loan rate i that would allow you to break even if you had to borrow the funds to invest, and you waited until the end of the horizon to repay the loan, while accumulating cash inflows in a fund earning j:

$$\sum_{t=0}^{n} A_t (1+j)^{n-t} = \sum_{t=0}^{n} B_t (1+i)^{n-t}$$

Example 7.6

Find the yield rate on the following investment if

(a) the reinvestment rate is 5%.



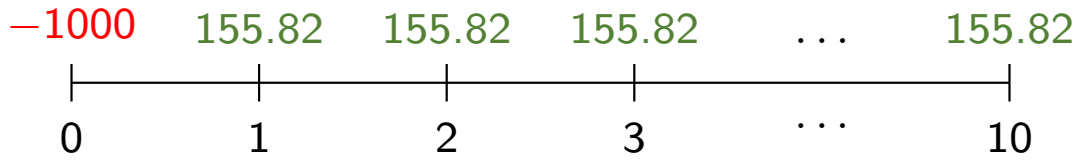
Example 7.6, continued

(b) the reinvestment rate is 7%.



Example 7.6, continued

(c) the reinvestment rate is 9%.



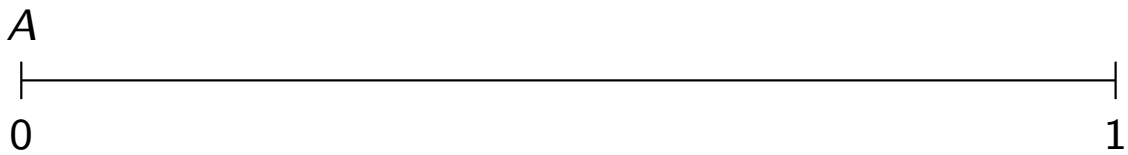
7.3 Dollar-weighted rates

Notation:

- A: Amount in fund at $t = 0$
- B: Amount in fund at $t = 1$
- I: Interest earned in the year
- n_t: New principal added at time t, $(0 \leq t \leq 1)$
- $n = \sum n_t$: total new principal added during year
- w_t: Principal withdrawn at time t, $(0 \leq t \leq 1)$
- $w = \sum w_t$: total withdrawals during year

$$B = A + n - w + I$$

Equation of value



If we assume compound interest during the year:

$$B = A(1+i) + \sum n_t(1+i)^{1-t} - \sum w_t(1+i)^{1-t}$$

We know that the interest rate i that satisfies this equation is the IRR. But solving this is cumbersome.

A simple approximation

If we assume simple interest during the year, the equation of value at $t = 1$ becomes

$$B = A(1+i) + \sum n_t \big(1 + i(1-t)\big) - \sum w_t \big(1 + i(1-t)\big)$$

$$B = A + n - w + iA + \sum n_t\, i(1-t) - \sum w_t\, i(1-t)$$

Dollar-weighted rate of interest

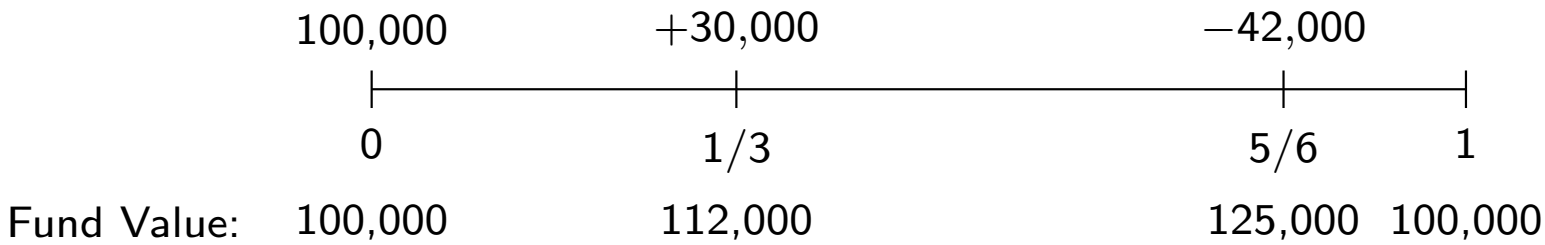
The **dollar-weighted rate of interest** is an approximation to the actual rate of interest earned, based on the simple interest assumption:

$$i \approx \frac{I}{A + \sum n_t(1-t) - \sum w_t(1-t)}$$

This is usually a good approximation, especially if the n_t and w_t are small compared to A.

Example 7.7

Consider the following pension fund:



Calculate the dollar-weighted rate of interest.

A further simplification

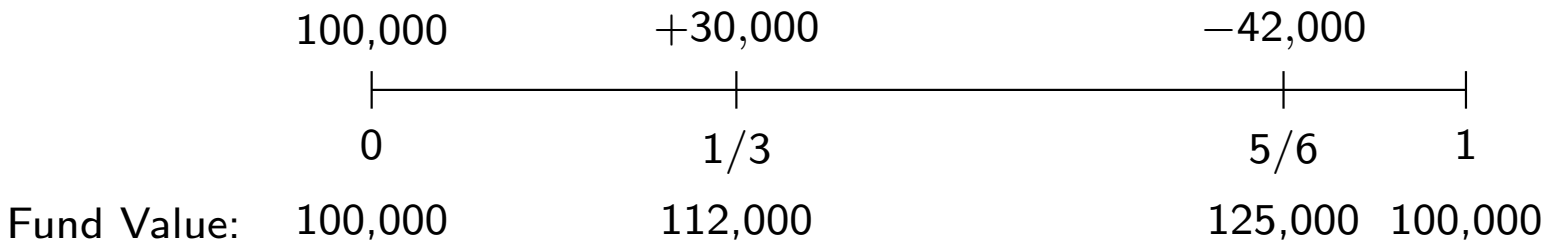
If we assume that the cash flows n and w occur uniformly throughout the year (midyear on average)

$$i \approx \frac{I}{A + 0.5n - 0.5w}$$

This is widely used in practice, but provides a poor approximation if the timing of the cash flows differs significantly from the assumption.

Example 7.7, continued

Consider the following pension fund:



Recalculate the dollar-weighted rate of interest using the midyear assumption.

7.4 Time-weighted rates

1. Divide the period under consideration into m successive subintervals each time a deposit or withdrawal is made.
2. Denote subinterval k as $[t_{k-1}, t_k]$
3. Calculate the fund balance B_t at the end of each subinterval, just before each cash flow
4. Calculate the rate of return during subinterval k as

$$1 + j_k = \frac{B_{t_k}}{B_{t_{k-1}} + n_{t_{k-1}} - w_{t_{k-1}}}$$

Time-weighted rates

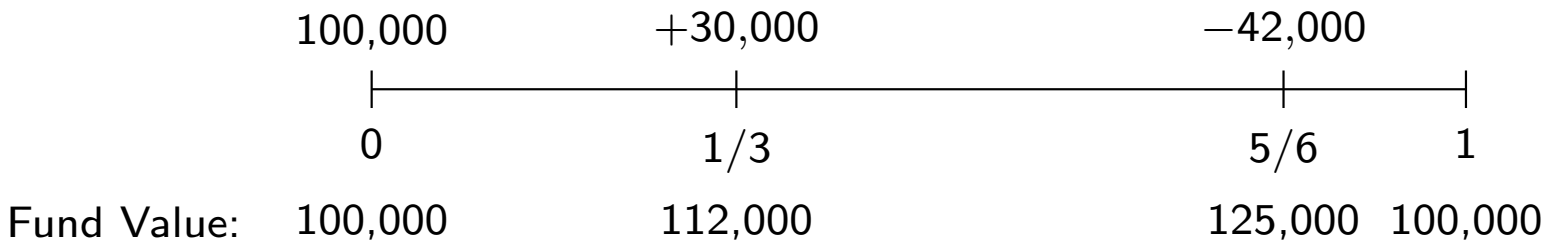
5. Then the time-weighted rate of interest, i, earned over the entire year is calculated as

$$1 + i = \prod_{k=1}^{m} (1 + j_k)$$

Note: Time-weighted rates are not consistent with compound interest, but they are usually better indicators of the performance (or effectiveness) of the investment manager.

Example 7.7, continued

Consider the following pension fund:



Calculate the time-weighted rate of interest for the fund.

7.5 Other securities

Debt versus equity in the corporate structure:

- Debt holders are called creditors; they always get paid unless the borrower defaults on the loan
  - Secured creditors (have lien on assets)
  - Unsecured creditors (no lien)
- Equity holders (aka shareholders) are entitled to residual profits, after everyone else is paid
  - Periodic disbursements of profits to shareholders are called dividends

Different types of securities

- Equity securities:
  - Preferred stock: fixed dividends, paid before common stock dividends
  - Common stock: dividend varies with profits
- Debt securities
  - Bonds (Chapter 6)

Dividend discount model

- Notation
  - D: amount of next dividend payment
  - i: investor's desired rate of return (yield)
  - g: growth rate of the dividend
  - P: price of the stock
- Stock price according to dividend discount model

$$P = \frac{D}{1+i} + \frac{D(1+g)}{(1+i)^2} + \frac{D(1+g)^2}{(1+i)^3} + \cdots = \frac{D}{i-g}$$

Example 7.8

The earnings of a corporation are expected to increase 3% per year. The current earnings on a common stock are \$5 per share. The corporation pays 60% of its earnings as dividends. Find the price per share that will yield 4.5%.